

ARBER SHALA

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EDUCATION

University of Alberta - Bachelor of Science with Specialization in Computer Science 09/2020 - 04/2025

Relevant Coursework: Information Retrieval, Machine Learning, Software Engineering, Numerical Methods

EXPERIENCE

University of Alberta: Cmput 401 Product Management Capstone Edmonton, Alberta

Software Developer 09/2024 - 12/2024

- Built a Django REST framework web application in Python for a pharmacology professor, enabling easy access to archive question data and streamlining the exam creation process
- Designed a well-organized database to effectively manage hundreds of questions and dozens of exams
- Managed project workflow with teammates using GitHub for version control and documentation, maintaining CI/CD pipelines to ensure continuous integration and reduce merge conflicts
- Created automated test suites in Python to verify code functionality, increasing confidence in code reliability before deployment
- Served as Scrum Master, applying Agile Software practices to keep the project on schedule
- [Github Pages Link](#)

University of Alberta: Cmput 469 AI Capstone Edmonton, Alberta

Student Researcher 01/2025 - 04/2025

- Supervised by Dr. Russ Greiner
- Explored how two backpropagation methods, Real Time Recurrent Learning (RTRL) and Backpropagation Through Time (BPTT) compare in performance when training Streaming Deep Reinforcement Learning Agents in partially observable environments
- Used the scientific method to design and conduct experiments to collect statistically significant data that reveals distinct performance trade-offs between the two methods
- Produced a research paper containing background on RTRL and BPTT, the design of our experiments, and the results produced from each experiment
- [Link to paper](#)

PROJECTS

Flight Price Data Manager | Python, PostgreSQL, Databricks, Golang 05/2025 - present

- Developed a web scraper to scrape flight price data from various sources.
- Designed an ELT pipeline using the medallion architecture to digest data into a postgresSQL database.
- Developing an accessible user interface to conveniently access flight price data.
- [Github Link](#)

High Volume ELT Pipeline and Data Warehouse | SQL, MSSQL 10/2025 - 11/2025

- Designed an ELT pipeline using the medallion architecture that ingests thousands of rows of structured source data from CSV files and processes the data into a business-ready data warehouse hosted on MSSQL
- Developed well-documented stored procedures with in-built error checking to execute complex SQL queries that populate each layer of the data warehouse with data
- [Github Link](#)

TECHNICAL SKILLS

Languages: Python, SQL, Java , Golang, HTML

Frameworks and Libraries: Django, Flask, Numpy, Matplotlib, Pandas, Pytorch, scikit learn, statsmodels

Tools: Git, Docker, MSSQL, RESTful API, Databricks, Power BI, JSON, Excel, Apache, Google Colab, Azure